Exploring the Landscape of Tourism Knowledge Graphs: A Systematic Literature Review

Hendrik Hendrik[1,2], Sri Suning Kusumawardani[1], Adhistya Erna Permanasari[1]
[1]Department of Electrical and Information Engineering, Universitas Gadjah Mada, Yogyakarta, Indonesia
[2]Department of Informatics, Universitas Islam Indonesia, Yogyakarta, Indonesia
e-mail: {hendrik1880@mail.ugm.ac.id, hendrik@uii.ac.id}, {suning | adhistya}@ugm.ac.id

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***Abstract*—Knowledge graphs (KGs) are important tools for organizing and leveraging machine-readable knowledge across various domains. This paper presents a systematic literature review examining the state of research on tourism knowledge graphs (TKGs). Our review followed Kitchenham guidelines and developed four research questions to accomplish the objectives of the study. We analyzed 35 relevant studies from 151 English-language publications published between 2013 and 2023 to identify trends in TKG construction, application, and evaluation. Our findings revealed that TKGs are predominantly built using graph data models such as RDF and LPG, stored in graph databases such as GraphDB and Neo4J, and extracted from diverse tourism data sources. The main applications include tourism recommender systems, question-answering systems, conversational assistants, and data analytics. European TKG projects prioritize semantic interoperability, while Asian projects focus on performance optimization. The TKG quality is evaluated primarily through metrics of correctness and completeness or through case study applications. However, our review identified several key gaps: limited scope of applications focusing primarily on tourist-centric use cases, lack of comprehensive construction methodologies, insufficient evaluation across multiple quality dimensions, and underutilization of TKGs for sustainable and responsible tourism practices. We propose directions for future research to address these limitations, including developing TKGs to support sustainable tourism practices, integrating TKGs with emerging technologies, expanding applications to benefit a wider range of stakeholders, and enhancing evaluation methodologies. This study contributes to the advancement of intelligent tourism systems by providing a comprehensive overview of the TKG landscape and guiding future research efforts.**

***Index Terms*—Intelligent Tourism, Knowledge Graphs, Artificial Intelligence, Semantic Web, Knowledge-based system, Systematic Literature Review**

I. Introduction

Knowledge Graphs (KGs) are powerful tools for organizing and leveraging machine-readable knowledge across various domains [1], [2]. Since Google's popularization of the term in 2012, KGs have found applications in fields ranging from education to healthcare and finance [3], [4]. At their core, KGs represent structured, interrelated datasets with semantic annotations that provide rich contextual information [5].

The tourism industry, characterized by its information-intensive nature and rapid digitalization, presents a compelling case for KG application. Studying Knowledge Graphs in tourism and hospitality is crucial due to these sectors' unique characteristics: vast amounts of heterogeneous data, complex stakeholder relationships, and rapidly changing consumer preferences. Tourism Knowledge Graphs (TKGs) offer the potential to revolutionize how tourism information is organized, accessed, and utilized [6]. KGs can integrate diverse information sources (e.g., attractions, accommodations, transportation, reviews) into a cohesive, easily queryable format, enabling more sophisticated applications like personalized travel planning, real-time destination management, and cross-sector trend analysis.

The urgency of TKG research is underscored by several factors: increasing demand for personalized travel services, the need to address information overload, growing complexity of tourism ecosystems, and the rising importance of data-driven decision-making in destination management. Key beneficiaries include: (1) tourists, who receive more personalized and context-aware recommendations; (2) tourism businesses, which can make data-driven decisions and offer tailored services; (3) destination management organizations, which can better understand and manage visitor flows and experiences; and (4) policymakers, who can leverage comprehensive insights for sustainable tourism development. By enhancing information accessibility and facilitating intelligent data utilization, TKGs have the potential to transform the entire tourism ecosystem.

While the application of KGs in tourism has been growing, there remains significant potential for further development and broader implementation [7]. Despite the promising prospects, the field of TKG research is still evolving, with various challenges and opportunities yet to be fully explored.

This paper presents a systematic literature review examining the state of research on TKGs. Our objectives are to analyze trends in TKG construction, application, and evaluation; identify key challenges and opportunities in TKG development; and propose directions for future research to address current limitations and expand the impact of TKGs. By providing a comprehensive overview of the TKG landscape, this study aims to guide future research efforts and contribute to the advancement of intelligent tourism systems.

II. Related Literature Reviews

The field of knowledge graph (KG) research is extensive, covering foundational overviews and specific applications across numerous domains. Existing literature reviews on KGs reveal varying approaches and focuses.

Hogan et al. [8] provide a comprehensive introduction to KGs, emphasizing their technical foundations and broad applications. In contrast, Ji et al. [9] take a more specialized approach, categorizing KG research into critical areas such as knowledge representation learning and acquisition.

While these reviews offer valuable insights into KGs generally, they differ in their treatment of domain-specific applications. Abu-Salih [10] surveys domain-specific KGs, touching on various fields including healthcare and finance, but does not delve deeply into tourism applications.

Our review uniquely focuses on the emerging niche of Tourism Knowledge Graphs (TKGs), an area where the challenges and opportunities of applying KGs are still being explored. This sets our work apart from broader KG discussions and fills a critical gap in the existing literature review landscape. By emphasizing the untapped potential of TKGs in enhancing personalized and intelligent tourism services, our review establishes a narrative that aligns with and expands upon insights from foundational KG literature, setting a new course for TKG research.

Unlike general KG literature reviews, which cover a wide array of applications and technological discussions, our review targets explicitly the intersection of KGs with the tourism sector. This involves a detailed examination of TKG construction methodologies, applications, and potential benefits to the tourism industry.

Furthermore, our review emphasizes the significance of advanced evaluative metrics and methodological innovations tailored to the tourism domain. While broader KG literature discusses aspects such as KG quality control, our review highlights explicitly how these considerations apply to TKGs, stressing the importance of accuracy, reliability, and utility in enhancing tourism experiences.

III. Methods

We conducted a systematic literature review to assess state-of-the-art tourism knowledge graph research and potential future research challenges. This section provides a framework for conducting a systematic literature review according to the recommendations in [11].

A. Formulating Research Questions

We formulated the following four research questions to achieve the objective of this study:

Research Question 1 (RQ1): What are the characteristics of the knowledge graph on Tourism Knowledge Graph?

Research Question 2 (RQ2): How are TKGs constructed?

Research Question 3 (RQ3): To what extent have TKGs been used?

Research Question 4 (RQ4: How are the TKGs evaluated?

B. Defining Search Strategy

We defined a search strategy for this review to gather relevant papers from the Scopus database. We implemented several parameters in our search approach, including keywords, search techniques, publication timeframe, publication type, publication source, and publication language. Each parameter is described in detail below.

Our primary search term was "tourism knowledge graph." We implemented a search query on titles, abstracts, and keywords. We included papers published between 2013 and 2023 to capture the most recent decade of developments in TKG research, aligning with the popularization of Knowledge Graphs by Google in 2012. This timeframe allows us to trace the evolution of TKG research from its inception to the current state of the art.

We selected Scopus as our primary database due to its comprehensive coverage of peer-reviewed literature in computer science and tourism, quality assurance of peer-reviewed content, advanced search features, and interdisciplinary scope. However, we acknowledge that using a single database is a limitation of our study.

Note: While our initial search was conducted in 2023, the publication process has extended into 2024. We acknowledge this timing discrepancy and its potential impact on our findings.

C. Selecting Articles

We collected data from July 2023 and received 151 documents after implementing the search technique. We then used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework to screen the results [12]. The papers were screened in four stages, as shown in Figure 1. We explained each step of the screening procedure.

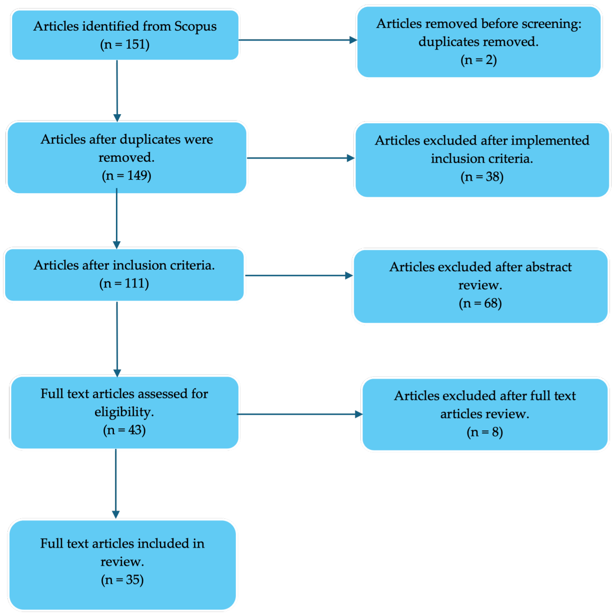

Fig. 1: Diagrammatic representation of the PRISMA selection process for articles

In the first stage, duplicates were eliminated. Consequently, two articles were excluded. Hence, during this phase, the results decreased from 151 to 149. During the second phase, we used inclusion criteria to reject documents irrelevant to our study's goals. Our review's inclusion criteria were as follows: First, the articles were published between 2013 and 2023 in the publication timeframe. Second, they must be peer-reviewed journal publications or conference proceedings published in English. The following are all the search keywords that were used after implementing the inclusion criteria. As a result, we received 111 documents that met our inclusion criteria.

TITLE-ABS-KEY (tourism OR travel AND "Knowledge Graph") AND PUBYEAR > 2013 AND PUBYEAR < 2023 AND (LIMIT-TO (DOCTYPE, "cp") OR LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English"))

In the third phase, we applied the exclusion criteria by examining document abstracts. "Review papers" and "publications that do not discuss tourist knowledge graph development or application" are among the exclusion criteria used. In this phase, we removed 68 studies that met the exclusion criteria. Finally, during the screening phase, we read the full-text publications. We implemented the same exclusion criteria and obtained 35 papers for the review process.

D. Reviewing Articles

We employed a semi-automated approach to extract relevant details from the final 35 articles. An AI-based assistant, Claude-2, was used to analyze the following information:

1) Knowledge graph characteristics: graph data models, databases, and integrated knowledge sources.
2) Construction approaches: extraction and fusion techniques, data storage methods, and adopted frameworks.
3) Applications: types of systems or use cases where the TKG was applied.
4) Evaluation methodology: specific metrics, quality dimensions, and verification processes.

To ensure accuracy, our team manually verified key elements of the AI's analysis against the original papers. This human oversight confirmed that Claude-2 correctly captured all relevant metadata.

We then consolidated these details into a spreadsheet for trend analysis across studies. By aggregating evidence from the sample, overarching themes emerged around prevalent technologies, applications, and evaluation practices. This systematic examination guided our insights into the current research landscape and identified areas requiring further attention in TKG development.

IV. Results

Before discussing our study's findings, we will discuss the demographic analysis of the examined papers. First, as we can see in Figure 2, conference papers dominated the distribution, with 69% of the total publications. In contrast, journal articles comprise 31%. This trend aligns with a common practice in computer science, where researchers prefer to publish in conference proceedings over journals. The preference is due to is due to their predictability. Conferences have well-defined submission deadlines, author feedback, and notifications [13], [14]. This makes the conference publication process more foreseeable than for journals, which often have less rigid timeframes. In addition, the total citations of the reviewed papers published in proceedings were almost three times higher than those published in journals. Moreover, proceedings have higher average citations per document than journal articles (Figure 3). Our finding verifies the research of Vrettas & Sanderson, which concluded that the publication practice in CS differs significantly from that of other academic fields [15].

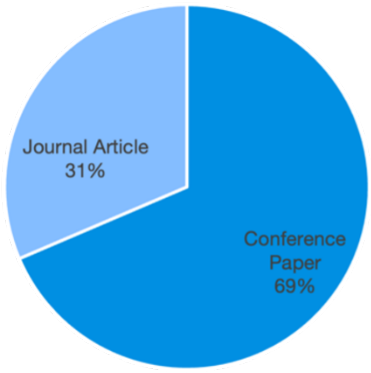

Fig. 2: Distribution of publications between journal articles and conference papers.

Furthermore, Figure 4 shows a visualization of publication trends. In the initial year of 2016, there were only two publications, indicating that this was a newly emerging research area. Over the next few years, interest in the topic has fluctuated, peaking at nine publications in 2020 and 2021. This spike suggests that tourism knowledge graphs have become a popular research topic in recent years. Subsequently, the number of publications declined to five by 2022 and 2023, respectively.

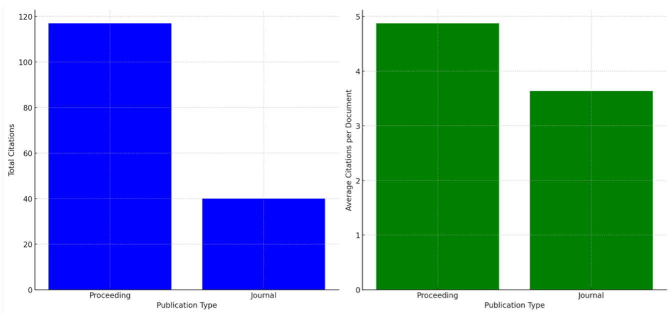

Fig. 3: Comparative visualisation of total and average citations for proceedings and journals.

Meanwhile, based on the object locations of the study, we found that the research of TKG currently focuses on Asia and Europe. A striking observation is China's dominance, with 19 publications, which leads the research on TKG. Vietnam, Spain, Italy, and Germany, exhibit moderate contributions, with publications ranging between 2 to 3 publications each. Several additional countries, including Austria, Thailand, the USA, France, and Malaysia, publish one publication each (Figure 5). More broadly, the overall Knowledge Graph publication data analysis in the Scopus scholarly database shows that China has emerged as a global leader in the knowledge graph research field.

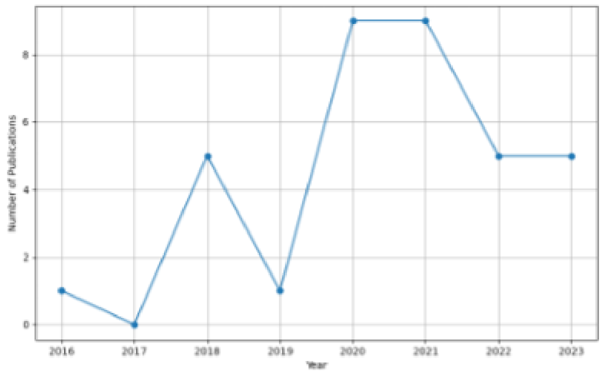

Fig. 4: Annual trend of publications related to tourism knowledge graphs from 2016 to 2023

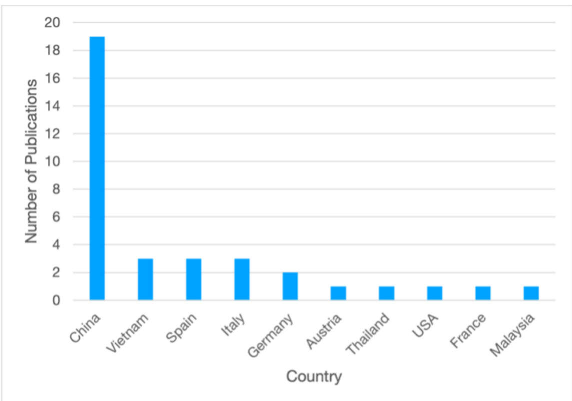

Fig. 5: Tourism knowledge graph publications by country

A. RQ1. Characteristics of the knowledge graph on Tourism Knowledge Graph

We determine the knowledge graph features of TKG based on three technological aspects: the graph data model, graph data store, and knowledge resource type. The followings are the outcomes and discussions for each character:

1) Graph Data Model: The choice of data model for constructing Tourism Knowledge Graphs (TKGs) significantly impacts information representation, integration, and querying capabilities. Our review reveals distinct patterns in data model selection that align with regional preferences and project-specific needs.

European TKG projects predominantly use the Resource Description Framework (RDF) model for its robust semantic capabilities and standardization, facilitating information exchange and interoperability in the diverse tourism industry [16], [17]. In contrast, Asian projects prefer the Labeled Property Graph (LPG) model for its storage efficiency and faster query response times, supporting dynamic, user-facing tourism services [18], [19].

These findings indicate that data model choice is influenced by geographical context and specific TKG requirements. European TKGs prioritize semantic interoperability, while Asian TKGs focus on performance optimization. This tailored approach underscores the need for a nuanced understanding of how technical choices intersect with tourism industry needs, highlighting the potential for future research into hybrid or extended data models to meet the full spectrum of intelligent tourism requirements.

2) Graph Data Store: In the domain of Tourism Knowledge Graphs (TKGs), the choice of graph data store is crucial for performance, scalability, and the richness of semantic queries. This study observed a preference for specific graph databases that aligned with TKGs' aims and technical requirements.

Ontotext GraphDB has been noted for its semantic web and linked data capabilities, making it a favorable choice for projects where rich semantic queries and W3C compliance are pivotal [20], [21]. These features are particularly relevant for TKGs, where semantically enriched content is necessary to provide detailed and context-aware information to users. The use of GraphDB in European TKGs signifies an emphasis on leveraging the Semantic Web's potential for enhancing tourist experiences.

Virtuoso's dual function as a knowledge graph store and a data-query interface underlines its robustness for large-scale TKGs [7]. Its adaptability across SQL, SPARQL, and GraphQL standards makes it an excellent tool for integrating heterogeneous data sources—a common scenario in the tourism sector where diverse data sets must be harmonized.

On the other hand, Neo4J's popularity in Asian TKG projects can be attributed to its native graph processing capabilities and ease of use, which are advantageous for developers working on innovative tourism applications requiring rapid development cycles [18], [19]. Its scalability and active community make Neo4J a practical choice for growing TKGs that must adapt to changing data and user demands.

These patterns suggest that the choice of graph data store for TKGs is influenced not only by the technologies' inherent properties but also by the regional focus of TKG research and the specific demands of tourism-related use cases. European TKGs tend to prioritize Semantic Web compatibility, while Asian TKGs emphasize performance and ease of development, reflecting differing strategic approaches to building intelligent tourism systems.

3) Type of Knowledge Sources: Selecting knowledge sources is critical in constructing TKGs, as they underpin the graph's depth, breadth, and utility. Our study categorizes these sources along with their specific roles and contributions to the development of TKGs (Table I).

B. RQ2. Construction of Tourism Knowledge Graph

The construction of a KG involves several activities that vary depending on its specific characteristics, such as the graph model, graph store employed, and types of knowledge resources and data (e.g., structured, semi-structured, or unstructured). However, after reviewing TKG research articles, we discovered no generalized framework or methodology to develop TKG. Moreover, many evaluated papers on constructing tourism knowledge graphs (TKGs) have insufficient coverage of the overall construction approach or important construction tasks [17], [24], [25], [27], [29], [32]. Some studies have focused only on essential construction tasks such as knowledge extraction [6], [26], [33], design [17], or fusion [34]. This lack of clarity and specificity can make it challenging for other researchers to reproduce or expand the methods stated in these studies. In addition, the absence of publicly accessible TKGs for review or future development inhibits knowledge exchange, translation, and reuse and may restrict the potential effect of the study of TKG.

Future studies may benefit from adopting a more holistic approach that considers all stages of TKG development, including knowledge extraction, ontology design, and knowledge fusion, to solve these challenges and improve TKG construction research. Another option would be to design a framework or set of principles for TKG building that considers the various elements that influence the success of a TKG, such as the size and complexity of the domain, availability of data sources, and the requirements of various stakeholders. To make it easier for researchers to build upon previous work and contribute to building a more comprehensive and useable TKG, a second alternative would be to investigate the use of open-source tools and resources for TKG construction. It would also benefit researchers to make their TKGs publicly available for evaluation and further development to facilitate knowledge-sharing and collaboration within the research community.

C. RQ3. Applications of Tourism Knowledge Graph

Tourism knowledge graphs (TKGs) have found diverse applications, primarily catering to the needs of tourists but also benefiting other stakeholders like destination marketing organizations and travel operators [17], [20]–[22], [28]. A prominent application lies in recommender systems, which leverage TKGs to suggest travel destinations, accommodations, activities, restaurants, and attractions aligning with users' preferences, histories, and contextual factors. Numerous studies have proposed recommender models built upon TKGs, employing techniques such as convolutional neural networks [35], [36], knowledge graph feature learning [37], and node embeddings [32]. Personalized recommenders tailoring suggestions based on user interest models derived from TKG entity properties have also gained traction [19], [22], [23], [38]–[41].

TKGs further facilitate question-answering (Q&A) systems capable of responding to queries about travel locales, accommodations, activities, local customs, and cultural nuances. Approaches include integrating TKGs with deep learning [18], leveraging BERT+BiLSTM+CRF for entity extraction [31], utilizing CNNs to map questions to KG attributes [42], and employing semi-automatic knowledge acquisition techniques [43]. Conversational assistants, powered by natural language processing and TKGs, provide real-time travel information and support, assisting with tasks like trip planning, reservations, and recommendations [20], [38].

Moreover, TKGs enable data analysis to uncover insights and patterns from large tourism datasets, such as accommodation price trends [20] and multidomain analyses combining tourist and climate data [44]. Knowledge application platforms harnessing TKGs offer intelligent browsing and search experiences, personalized and semantically relevant information, open data access, and multimodal knowledge sharing [7], [17], [34].

While the majority of TKG applications prioritize tourists' needs, a developing stream of research explores their potential for benefiting other tourism stakeholders like destination marketing organizations and travel operators. As TKGs continue to evolve, their impact on enhancing the tourism industry's efficiency, personalization, and knowledge dissemination is poised to grow.

D. RQ4. Evaluation of Tourism Knowledge Graph

Evaluating TKG quality is crucial for ensuring reliability and effectiveness. Two main approaches were identified:

1) Gold standard/metrics-based approach (36% of studies):
   a) Uses precision, recall, and F1-score [6], [19], [31], [45]
   b) Some studies incorporated accuracy [7] or AUC and top-N precision [36]
   c) NER and relation extraction (RE) [45], and NER and entity alignment (EA) [33]
2) Case study/application-based approach (72% of studies) Assesses TKG usability in specific tasks like recommender systems and question-answering

Some studies combined both approaches [7], [19], [31], [45]. However, no manual or domain expert-based evaluations were found. Most evaluations focused on accuracy and completeness [6], [7], [19], [24], [31], [36], [45], with only [46] considering both. Other quality dimensions like consistency, timeliness, trustworthiness, and availability were neglected.

The quality of a knowledge graph (KG) is inherently tied to its construction process, necessitating quality control measures at various stages, including source selection, extraction, and fusion [47]. TKG research has primarily focused on knowledge extraction, likely due to its importance in ensuring graph correctness and completeness. However, considering the full range of factors influencing KG quality is essential. Broadening research to encompass source selection and fusion processes could further enhance TKG quality and applicability in tourism.

TABLE I: Classification of TKGs Knowledge Sources

Type of KS	Examples	Usage	Implications	References.
Open Knowledge Graph	DBpedia, Wikidata, GeoNames, wikitravel	Provide foundational knowledge layers for TKGs; basic facts about locations and entities	Accelerates TKG development; focus on domain-specific enrichment	[16], [18], [22]
Online Travel Agency	Booking.com, Airbnb, Ctrip	Offer structured data for real-time insights into travel services	Enhances user-centricity of TKGs; more dynamic and responsive to market	[23]–[25]
Tourism Board Databases	Spain.info, Tourism Malaysia	Provide curated, authoritative destination information	Ensures credibility and reliability of TKGs	[20], [22], [24]
Collaborative Travel Websites	TripAdvisor, Mafengwo	Offer user-generated reviews and experiences	Enriches TKGs with qualitative insights; more nuanced tourist experiences	[26]–[28]
Destination Websites	Beijing Tourism Network, Mount Tai official website	Provide localized destination content	Enhances personalization and context of TKGs	[24], [27], [29]
Hybrid with Other Data Types	Flickr, Climatiq API	Combine tourism info with data such as weather forecasts	Expands scope and utility of TKGs; supports innovative applications	[27], [30], [31]

V. Discussion

Our systematic review revealed a diverse range of strategies and methodologies employed for the construction and application of Tourism Knowledge Graphs (TKGs). We found a trend in the tourism industry to increase the use of Tourism Knowledge Graphs (TKGs) to enhance the personalization and intelligence of tourism services. TKGs aims to facilitate a more enriched and user-centric travel experience.

The current scope of TKG applications is limited, with a predominant focus on tourists rather than other stakeholders in the tourism industry. TKGs have the potential to significantly enhance personalized and intelligent tourism services. This phenomenon demonstrates that researchers in the field of tourism knowledge graphs (TKG) continue to rely on conventional one-way communication models in their TKG applications [48]. This model prioritizes the needs and preferences of tourists while neglecting to offer feedback or support to other stakeholders. Hence, our research findings emphasize the considerable potential of broadening the utilization of TKGs to provide advantages to various stakeholders in the tourism industry, including destination marketing organizations (DMOs), official tourism boards, and travel operators. TKGs can potentially provide stakeholders with valuable insights derived from data analysis, enhanced monitoring capabilities, optimized recommendations, and innovative product offerings [49]. Moreover, engaging with these stakeholders is essential to achieving sustainable tourism, as it requires addressing multiple conflicting issues such as environmental preservation, economic health, guest satisfaction, and community well-being [50].

Moreover, from a methodological standpoint, most current research on TKGs focuses primarily on a specific set of techniques, such as entity extraction. However, there is a notable absence of a comprehensive methodology encompassing essential tasks like ontology design, fusion, curation, and quality control. The findings of our study are consistent with those of Hofer et al. [51], which provides a comprehensive outline of the essential procedures involved in constructing high-quality Knowledge Graphs (KGs). These procedures encompass various crucial aspects such as metadata management, ontology development, and quality assurance. The implementation of stricter, more transparent, and reusable construction methods has the potential to enhance effectiveness and facilitate knowledge dissemination. Accessibility is also a concern, as only a limited number of TKGs are available as open resources for validation and progress.

Regarding evaluation, the assessed TKGs primarily focused on correctness and completeness while neglecting other essential quality dimensions, such as consistency, timeliness, trustworthiness, and availability [52]. Several studies have discussed the importance of knowledge graph evaluation and proposed different approaches. For example, Wang & Wang offers a model for evaluating the trustworthiness of triples in industrial knowledge graphs, which improves error detection and graph completion tasks [53]. Moreover, Chen et al. proposed a knowledge graph evaluation framework with several quality dimensions [54]. The proposed dimensions include relevance, trustworthiness, understandability, and semantic accuracy. Meanwhile, existing methods for evaluating knowledge graph quality rely heavily on the graph's raw data, exposing internal information and focusing more on data level than ability level quality. However, Li et al. proposed a quality evaluation framework under incomplete information (QEII) to assess KGs at the ability level while protecting internal information [43].

While our review provides valuable insights into the current state of TKG research, it's important to acknowledge certain limitations of our study. Our use of a single database (Scopus) for the literature search, while providing comprehensive coverage and robust search features, may have potentially missed some relevant literature. This limitation could impact the comprehensiveness of our findings and should be considered when interpreting the results of this review.

Our review highlights the potential for future research on tourism knowledge graphs to address key industry challenges. We propose the following research directions:

1) Develop TKGs to support sustainable tourism practices, including environmental impact tracking, responsible tourism behavior facilitation, and overtourism management. This could involve modeling the relationship between tourist activities and their impact on destinations.
2) Explore the integration of TKGs with emerging technologies such as augmented reality, Internet of Things, and AI to create more immersive and context-aware tourism applications. This research should balance high-tech solutions with the high-touch nature of tourism services.
3) Expand TKG applications beyond tourist-centric use cases to benefit a wider range of stakeholders, including destination management organizations, travel operators, local communities, and policymakers. This could include developing TKGs for tourism policy modeling, impact assessment, and crisis management.
4) Investigate the development of TKGs for niche tourism markets and cross-cultural applications. This includes creating multilingual TKGs and those tailored to specific segments (e.g., cultural, religious, or eco-tourism), incorporating unique attributes, regulations, and preferences relevant to diverse tourist groups and cultural contexts.
5) Develop more comprehensive evaluation frameworks for TKGs that go beyond accuracy and completeness to include tourism-specific metrics such as user satisfaction, cultural sensitivity, and economic impact.

VI. Conclusion

This systematic review has mapped the rapidly evolving landscape of Tourism Knowledge Graph (TKG) research. Our findings reveal that while TKGs show great potential in enhancing personalized tourist experiences, several key gaps remain:

1) Limited scope of applications, primarily focusing on tourist-centric use cases.
2) Lack of comprehensive construction methodologies.
3) Insufficient evaluation across multiple quality dimensions.
4) Underutilization of TKGs for sustainable and responsible tourism practices.

Beyond theoretical contributions, our research yields several practical implications for the tourism industry:

1) Enhanced personalized recommendation systems for tourists
2) Data-driven decision making for destination management
3) Efficient information sharing among tourism stakeholders
4) Improved policy modeling and impact assessment for sustainable tourism development

Our findings suggest that researchers and practitioners should focus on developing more robust, specialized, and widely applicable TKGs that can address the complex challenges facing the tourism industry. As outlined in our discussion, future research directions should aim to bridge the identified gaps while maintaining a balance between technological innovation and practical industry needs.

By addressing these challenges, the field can unlock TKGs' full potential in driving sustainable, intelligent tourism development. This advancement will catalyze enriched experiences not just for tourists, but for all stakeholders in the multifaceted tourism industry, ensuring that TKGs continue to drive positive change in the tourism sector.

References

[1] C. Gutierrez and J. F. Sequeda, "Knowledge graphs," *Communications of the ACM*, vol. 64, pp. 96–104, 3 2021.

[2] Y. Liu, G. Wang, Y. Zhou, and Y. Liu, "Advanced technology evolution pathways of nanogenerators: A novel framework based on multi-source data and knowledge graph," *Nanomaterials*, vol. 12, 2022. [Online]. Available: https://doi.org/10.3390/nano12050838

[3] P. Chen, Y. Lu, V. W. Zheng, X. Chen, and B. Yang, "Knowedu: A system to construct knowledge graph for education," *IEEE Access*, vol. 6, pp. 31 553–31 563, 5 2018.

[4] F. Gong, M. Wang, H. Wang, S. Wang, and M. Liu, "Smr: Medical knowledge graph embedding for safe medicine recommendation," *Big Data Research*, vol. 23, p. 100174, 2 2021.

[5] J. Barrasa, A. Hodler, and J. Webber, *Knowledge Graphs*. O'Reilly, 2021.

[6] D. Xiao, N. Wang, J. Yu, C. Zhang, and J. Wu, "A practice of tourism knowledge graph construction based on heterogeneous information," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 12522 LNAI, pp. 159–173, 2020. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-63031-7_12

[7] J. Xie, Z. Dong, Q. Wen, H. Zhu, H. Jin, L. Hou, and J. Li, "Construction of multimodal chinese tourism knowledge graph," *Communications in Computer and Information Science*, vol. 1452 CCIS, pp. 16–29, 2021.

[8] A. Hogan, C. D. . Amato, G. De, C. Gutierrez, S. Neumaier, A. Polleres, S. M. Rashid, L. Schmelzeisen, . J. Emilio, L. Gayo, . A. Rula, S. Staab, A. Zimmermann, E. Blomqvist, M. Cochez, C. D'amato, G. D. Melo, S. Kirrane, J. Emilio, R. Navigli, A.-C. N. Ngomo, A. Rula, J. Sequeda, and A. Zim, "Knowledge graphs," *ACM Computing Surveys*, vol. 54, pp. 1–37, 2021. [Online]. Available: https://doi.org/10.1145/3447772

[9] S. Ji, S. Pan, E. Cambria, S. Member, P. Marttinen, P. S. Yu, and L. Fellow, "A survey on knowledge graphs: Representation, acquisition and applications," *IEEE Transactions On Neural Networks And Leraning Systems*, vol. 33, pp. 494–514, 2022. [Online]. Available: http://cyc.com

[10] B. Abu-Salih, "Domain-specific knowledge graphs: A survey," *Journal of Network and Computer Applications*, vol. 185, 7 2021.

[11] B. Kitchenham, O. P. Brereton, D. Budgen, M. Turner, J. Bailey, and S. Linkman, "Systematic literature reviews in software engineering – a systematic literature review," *Information and Software Technology*, vol. 51, pp. 7–15, 2009.

[12] M. J. Page, J. E. Mckenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. Mcdonald, L. A. Mcguinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, and D. Moher, "The prisma 2020 statement: an updated guideline for reporting systematic reviews," *International journal of surgery*, vol. 88, pp. 1–9, 2021. [Online]. Available: http://dx.doi.org/10.1136/bmj.n71

[13] M. Y. Vardi, "Publish <i>and</i> perish," *Communications of the ACM*, vol. 63, pp. 7–7, 12 2019.

[14] R. Klemiński, P. Kazienko, and T. Kajdanowicz, "Where should i publish? heterogeneous, networks-based prediction of paper's citation success," *Journal of Informetrics*, vol. 15, p. 101200, 8 2021.

[15] G. Vrettas and M. Sanderson, "Conferences versus journals in computer science," *Journal of the Association for Information Science and Technology*, vol. 66, pp. 2674–2684, 12 2015.

[16] A. Chessa, G. Fenu, E. Motta, F. Osborne, D. R. Recupero, A. Salatino, and L. Secchi, "Data-driven methodology for knowledge graph generation within the tourism domain," *IEEE Access*, vol. 11, pp. 67 567–67 599, 2023.

[17] R. Alonso-Maturana, E. Alvarado-Cortes, S. López-Sola, M. Martínez-Losa, and P. Hermoso-González, "La rioja turismo: The construction and exploitation of a queryable tourism knowledge graph," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 11153 LNCS, pp. 213–220, 2018.

[18] P. Do and H. Le, "Building a knowledge graph of vietnam tourism from text," in *Lecture Notes in Electrical Engineering*, 2021, pp. 1–13.

[19] X. Su, J. He, J. Ren, and J. Peng, "Personalized chinese tourism recommendation algorithm based on knowledge graph," *Applied Sciences (Switzerland)*, vol. 12, 2022.

[20] E. Kärle, U. Şimşek, O. Panasiuk, and D. Fensel, "Building an ecosystem for the tyrolean tourism knowledge graph," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 11153 LNCS, pp. 260–267, 2018.

[21] Şimşek U.; Angele K.; Kärle E.; Opdenplatz J.; Sommer D.; Umbrich J.; Fensel D., "Knowledge graph lifecycle: Building and maintaining knowledge graphs," in *CEUR Workshop Proc.*, 2021.

[22] D. A. T. R. D.-P. M. S. G.A., "Predicting your next trip: A knowledge graph-based multi-task learning approach for travel destination recommendation," in *CEUR Workshop Proc.*, 2021.

[23] C. Bin, T. Gu, Z. Jia, G. Zhu, and C. Xiao, "A neural multi-context modeling framework for personalized attraction recommendation," *Multimedia Tools and Applications*, vol. 79, pp. 14 951–14 979, 6 2020.

[24] P. Calleja, F. Priyatna, N. Mihindukulasooriya, and M. Rico, "Dbtravel: A tourism-oriented semantic graph," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 11153 LNCS, pp. 206–212, 2018. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-03056-8_19

[25] A. Cimmino, N. Mihindukulasooriya, F. Priyatna, and M. Rico, "Linked-fiestas: A knowledge graph to promote cultural tourism in spain," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 11153 LNCS, pp. 202–205, 2018. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-03056-8_18

[26] P. Do, T. Phan, H. Le, and B. Gupta, "Building a knowledge graph by using cross-lingual transfer method and distributed minie algorithm on apache spark," *Neural Computing and Applications*, vol. 34, pp. 8393–8409, 2022.

[27] L. Feng, "Design of tourism intelligent recommendation model of mount tai scenic area based on knowledge graph," in *Proceedings - 2020 International Conference on E-Commerce and Internet Technology, ECIT 2020*, 2020, pp. 241–244.

[28] J. Gao, P. Peng, F. Lu, C. Claramunt, and Y. Xu, "Towards travel recommendation interpretability: Disentangling tourist decision-making process via knowledge graph," *Information Processing Management*, vol. 60, p. 103369, 7 2023.

[29] W. Tao, Q. Zhou, Y. Zhao, and A. Yu, "A cross-field construction method of chinese tourism knowledge graph based on expasion and adjustment of entities," in *Proceedings of 2020 IEEE 5th Information Technology and Mechatronics Engineering Conference, ITOEC 2020*, 2020, pp. 211–215.

[30] A. Origlia, S. Rossi, S. D. Martino, F. Cutugno, and M. L. Chiacchio, "Multiple-source data collection and processing into a graph database supporting cultural heritage applications," *Journal on Computing and Cultural Heritage*, vol. 14, pp. 1–27, 12 2021.

[31] Y. Sui, "Question answering system based on tourism knowledge graph," *Journal of Physics: Conference Series*, vol. 1883, 2021.

[32] P. Yochum, L. Chang, T. Gu, M. Zhu, and W. Zhang, "Tourist attraction recommendation based on knowledge graph," *IFIP Advances in Information and Communication Technology*, vol. 538, pp. 80–85, 2018. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-00828-4_9

[33] M. Wu and H. Zhao, "Research on entity recognition and alignment methods in knowledge graph construction of multi-source tourism data," in *IMCEC 2021 - IEEE 4th Advanced Information Management, Communicates, Electronic and Automation Control Conference*, 2021, pp. 862–866.

[34] W. Zhang, H. Cao, F. Hao, L. Yang, M. Ahmad, and Y. Li, "The chinese knowledge graph on domain-tourism," *Lecture Notes in Electrical Engineering*, vol. 590, pp. 20–27, 2020. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-32-9244-4_3

[35] Q. Jiang, "A personalized attractions recommendation model based on tourism knowledge graph," in *Proceedings of SPIE - The International Society for Optical Engineering*, vol. 12129, 2021.

[36] J. Wu, X. Zhu, C. Zhang, and Z. Hu, "Event-centric tourism knowledge graph—a case study of hainan," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 12274 LNAI, pp. 3–15, 2020.

[37] F. Zeng and Y. Zheng, "Tourism recommendation system based on knowledge graph feature learning," *Proceedings of the 4th International Conference on Computing Methodologies and Communication, ICCMC 2020*, pp. 110–114, 3 2020.

[38] J. Lan, R. Shi, Y. Cao, and J. Lv, "Knowledge graph-based conversational recommender system in travel," in *2022 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 7 2022, pp. 1–8.

[39] C. Lu, P. Laublet, and M. Stankovic, "Travel attractions recommendation with knowledge graphs," in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 2016, pp. 416–431.

[40] C. Niu and W. Huang, "Tourism event knowledge graph for attractions recommendation," in *2023 8th International Conference on Cloud Computing and Big Data Analytics (ICCCBDA)*. IEEE, 4 2023, pp. 490–494.

[41] K.-S. She, S.-C. Haw, and L.-J. Chew, "Tourism recommender system utilising property graph ontology as knowledge base," in *Proceedings of the 12th International Conference on Computer Modeling and Simulation*. ACM, 6 2020, pp. 14–18.

[42] L. Yang, H. Cao, F. Hao, W. Zhang, and M. Ahmad, "Research on tourism question answering system based on xi'an tourism knowledge graph," in *Journal of Physics: Conference Series*, vol. 1616, 2020.

[43] J. Li, Z. Luo, H. Huang, and Z. Ding, "Towards knowledge-based tourism chinese question answering system," *Mathematics*, vol. 10, 2022.

[44] J. Wu, J. Pierse, F. Orlandi, D. O'Sullivan, and S. Dev, "Improving tourism analytics from climate data using knowledge graphs," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 16, pp. 2402–2412, 2023.

[45] X. Liang, H. Cao, and W. Zhang, "Knowledge extraction experiment based on tourism knowledge graph q a data set," in *Proceedings of 2020 IEEE International Conference on Power, Intelligent Computing and Systems, ICPICS 2020*, 2020, pp. 828–832.

[46] U. Simsek, E. Kärle, K. Angele, E. Huaman, J. Opdenplatz, D. Sommer, J. Umbrich, and D. Fensel, "A knowledge graph perspective on knowledge engineering," *SN Computer Science*, vol. 4, 2023.

[47] X. Wang, L. Chen, T. Ban, M. Usman, Y. Guan, S. Liu, T. Wu, and H. Chen, "Knowledge graph quality control: A survey," *Fundamental Research*, vol. 1, pp. 607–626, 9 2021.

[48] A. Bošnjak, G. Sladić, and N. Luburić, "Smart tourism destinations: Enhancing interconnection between stakeholders and tourist through information personalization and feedback system," in *7th International Conference on Information Society and Technology ICIST 2017*, 2017.

[49] M. A. Hamid, N. Rahmat, and A. S. A. Azmadi, "Stakeholders perception of smart tourism technology for tourism destination," *International Journal of Academic Research in Business and Social Sciences*, vol. 13, 4 2023.

[50] A. Hardy and L. Pearson, "Determining sustainable tourism in regions," *Sustainability*, vol. 8, p. 660, 7 2016.

[51] M. Hofer, D. Obraczka, A. Saeedi, H. Köpcke, and E. Rahm, "Construction of knowledge graphs: State and challenges," 2 2023. [Online]. Available: https://arxiv.org/abs/2302.11509v2

[52] L. Sebastian-Coleman, *Dimensions of Data Quality*. Elsevier, 2022, pp. 229–256.

[53] S. Wang, "Construction and analysis of tourism mapping based on data mining and apriori algorithm," in *Second International Conference on Algorithms, Microchips, and Network Applications (AMNA 2023)*, A. P. Muthuramalingam and K. Subramaniam, Eds. SPIE, 5 2023, p. 27.

[54] H. Chen, G. Cao, J. Chen, and J. Ding, "A practical framework for evaluating the quality of knowledge graph," *Communications in Computer and Information Science*, vol. 1134 CCIS, pp. 111–122, 2019.